P04: Makers Makin' It, Act II – The Sequel

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Description:

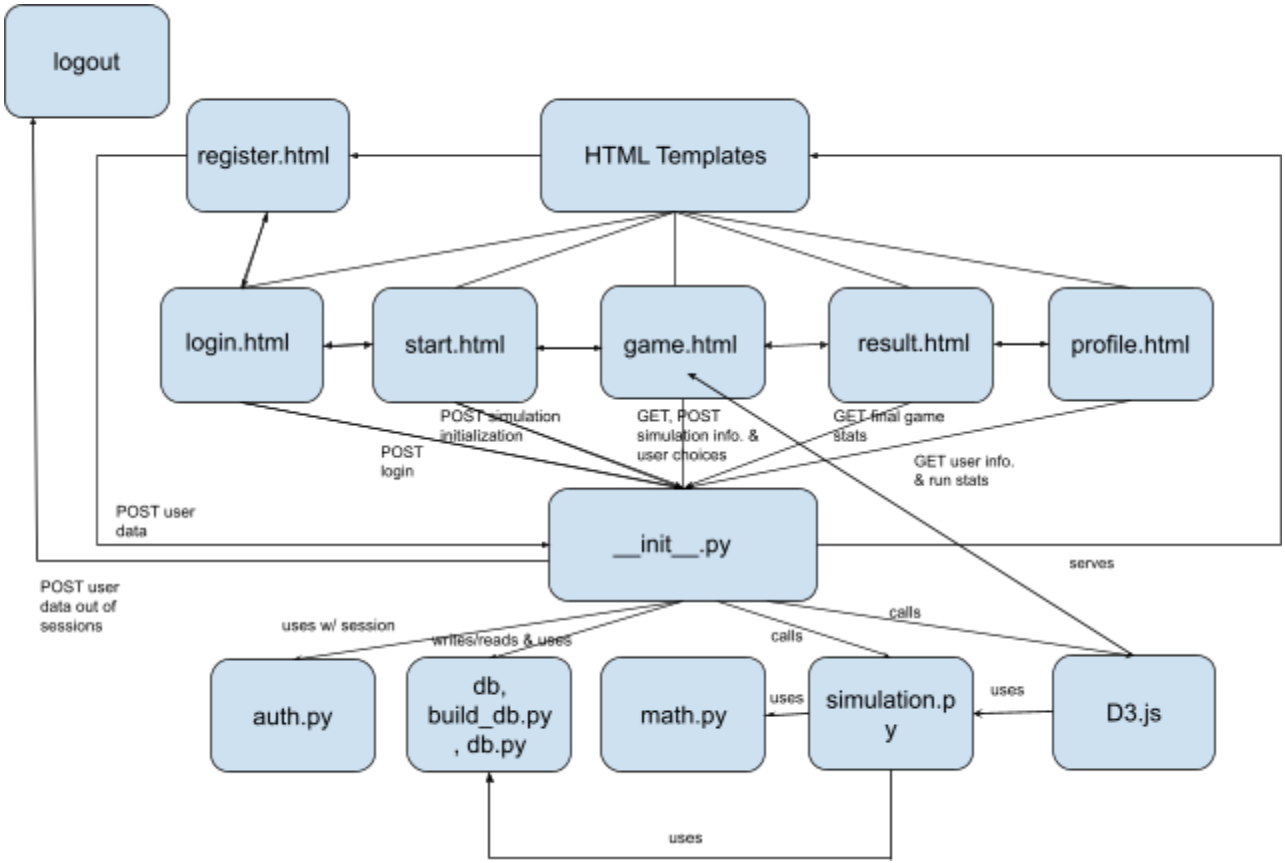
This project is an interactive, story-like Pandemic Simulator that incorporates both user data exploration and visualization to understand how different non-health factors and actions like population density, policy, and environmental and social vulnerability affect outbreaks Using Covid-19 data, the simulator has base outbreak infection and mortality rates that change based on user decisions and actions and are weighted based on correlations between the outbreak data and the factors, as well as some level of randomness.

Components:

- Webpage (frontend)
  - Tailwind for framework
  - Jinja2 HTML templates and user interface: login/registration, landing/start page, simulation page, logout, result and profile page
    - login/registration page automatically loaded if user not in session, otherwise redirect to landing, logout handles session removal, POST user authentication and session details to Flask app
    - landing/start page implements user interface, includes a short story introduction, start button, and initialization sliders/buttons to choose initial factors (population density, which decides location based on closest value in dataset, etc,), POST user input and simulation initialization to Flask app
    - simulation page implements user interface, top bar for narration, total stats, ticks/time, pause/stop/finish buttons, side bar for user decisions, container for maps/charts visualizing changing data with D3.js, GET and POST simulation information and updates with user input from and to Flask app
    - result page GET final stats, profile GET user information and final stats of each stimulation run (CHOPPING BLOCK)
- User Authentication (backend)
  - SQLite users table stores user_id PK and password pairs (SQLite instead of mongodb because most of our data is relational and standardized, not requiring a document)
  - Flask session and auth.py module w/ functions to create, check, and match user and password information

- Database (backend):
  - datasets: COVID-19 number infected and deaths by location over time, Social Vulnerability Index and population/population density by location, Stringency Index (tracks and rates countries' policy response to Covid-19) by location over time
    - scale: country, by state
  - 5 tables: users (user_id TEXT PK), country (country_id TEXT PK), stringency (strigency_id INT PK, country_id TEXT FK), covid_stats (stats_id INT PK, country_id TEXT FK), runs (run_id INT PK, user_id STRING FK, country_id TEXT FK) (CHOPPING BLOCK)
  - build_db.py module to initialize db, db.py module with data cleaning, processing, and query/insert/remove helper functions
- Additional Helper Modules (backend):
  - math.py for data and rate calculation (normalizing, averages, etc,) functions
  - simulation.py for maintaining and updating simulation states, stats, and narration using db.py and math.py
  - .js files for D3.js for map and graphs visualizing data over time, uses simulation.py
- Flask App (middleware):
  - handles routing, HTML templates, calls simulation.py and .js files
- Chopping Block: economics data tracking, result and profile page

Component Map:



Database Organization:

users		
TEXT	user_id	PK
TEXT	password	

countries		
TEXT	country_id	PK
INTEGER	vulnerability_index	
INTEGER	population	
FLOAT	population_density	

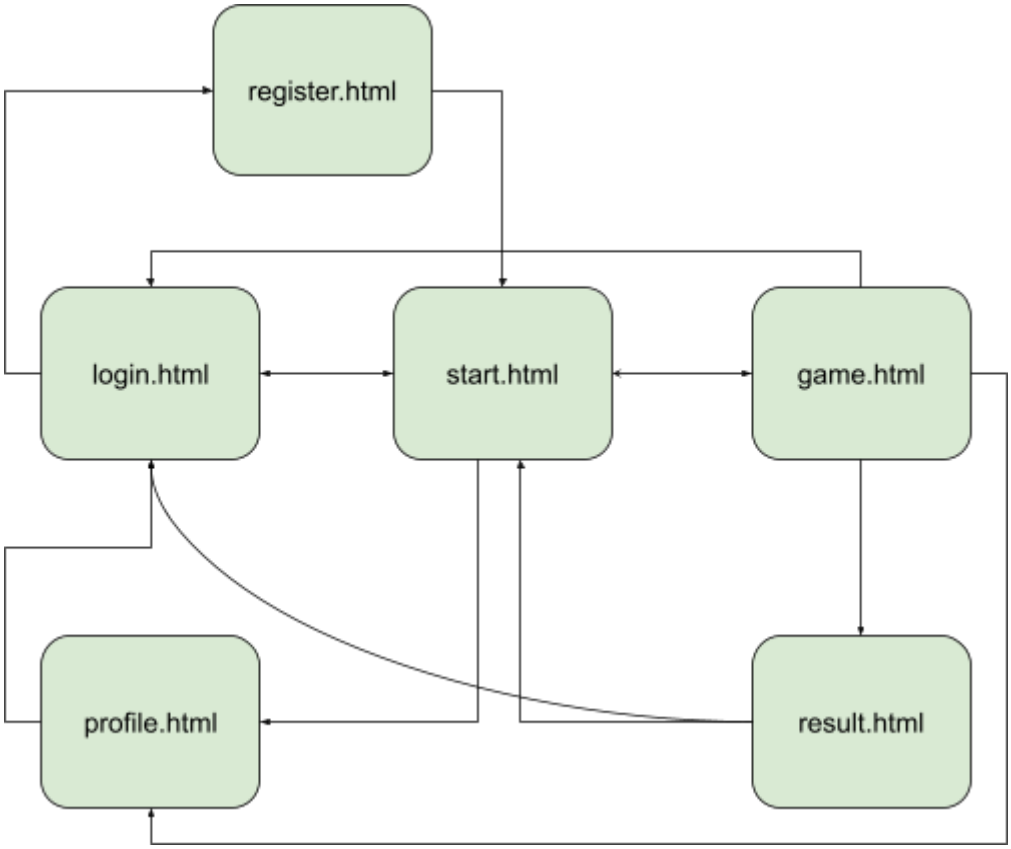
stringency		
INTEGER	stringency_id	PK
TEXT	country_id	FK
DATE	date	
FLOAT	stringency_index	

covid_stats		
INTEGER	stats_id	PK
TEXT	country_id	FK
DATE	date	
INTEGER	infected	
INTEGER	dead	

runs		
INTEGER	run_id	PK

TEXT	user_id	FK
TEXT	country_id	FK
INTEGER	total_infected	
INTEGER	total_dead	

Site Map:



Task Breakdown:

Sarah (PM):

- coordinate team check-ins, updates, and help/debug where needed
- build_db.py & db.py
- D3.js with joyce
- simulation.py with mottaqi

Joyce:

- Flask
- D3.js

Wesley:

- auth.py
- math.py

Mottaqi:

- html & css/framework
- simulation.py

FEF: Tailwind

- Tailwind is our chosen FEF for P04 as it allows for high degrees of customizability as opposed to its competitors such as Bootstrap and Foundation. While its selection of usable component samples is decidedly limited (with a majority being unfortunately locked behind a paywall), we will not likely require anything complicated in regards to CSS in this project.
- Specific tools we will be pulling from this framework include its flex-grid, color and size classes, as well as possible animations (likely purely for aesthetics as animations involved with data visualization will be handled with D3.js)

APIs:

- None!

Data Visualization Library: D3.js

- We will likely be choosing to use D3.js to generate the majority (if not all) of our graphs. Though its sample code is a lot harder to navigate than those offered by Chart.js (and also a lot longer), it allows for the visualization of data in much more complex ways including maps (world, stars, etc.), trees, color gradations, and the like. For our project particularly which has changing data and rates, the dynamic visualization of D3.js is a significant reason why we chose it. On the other hand, Chart.js has a much more simplistic appearance which we believe will be unable to deliver the effect we are looking for.